WRITING SAMPLE

Ruins in Use: Resilience and the Adaptive Transformation of War-Damaged Infrastructure in Berlin

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Abstract. Few European cities carry the physical traces of the Second World War as visibly as Berlin. Bombed churches, shelled museums, concrete bunkers and the fragments of demolished railway stations remain part of its everyday landscape. This paper asks how such war-damaged structures have been transformed rather than erased, and what these transformations reveal about the resilience of urban heritage. Drawing on conservation theory, from Riegl's notion of age value to the Venice and Burra Charters, and on the literature on adaptive reuse, it proposes four modes of transformation that are visible in Berlin: the conserved fragment, integrated repair, programmatic conversion, and landscape absorption. Each mode is discussed through a representative case. The paper argues that resilience in heritage terms is best understood not as the recovery of an original form but as the capacity of a damaged structure to keep producing meaning while taking on new uses. It concludes by considering the tensions this creates, between memory and commodification and between legibility and repair, and what the Berlin experience offers to other cities managing damaged or threatened heritage.

Keywords: adaptive reuse; urban resilience; war damage; architectural conservation; memory; Berlin

1. Introduction

At the end of the Second World War, a large part of Berlin's building stock lay damaged or destroyed. The decades that followed were shaped by the practical demands of clearing rubble and rehousing a population, and then by the political division of the city. In both halves, decisions about what to demolish, what to rebuild and what to leave as it was were rarely neutral. They expressed changing attitudes toward the past and competing visions of the city's future (Diefendorf 1993; Ladd 1997).

Many damaged structures were cleared. A significant number, however, survived in altered form: as memorial ruins, as repaired buildings that openly show their wounds, as converted shelters, or as hidden cores beneath new landscapes. These survivals form the subject of this paper. Rather than treating them as isolated exceptions, the paper reads them as evidence of a broader capacity of the city to absorb damage and to turn it into something usable and meaningful.

The central question is therefore twofold. First, through which architectural strategies have war-damaged structures in Berlin been adapted to a changing cityscape? Second, what do these strategies suggest about how resilience can be understood in the field of built heritage? To address these questions, the paper first sets out a conceptual framework linking conservation theory, adaptive reuse and resilience. It then outlines the historical context of postwar Berlin, before presenting a typology of four modes of transformation illustrated by representative cases. A discussion of the tensions within these approaches and a short conclusion follow.

2. Conceptual Framework

2.1 The value of the damaged monument

Modern conservation theory has long recognized that the meaning of a historic building is not limited to its original design. Alois Riegl's essay on the modern cult of monuments (1903) distinguished between the historical value of a monument as a document of a particular moment and its *age value*, the emotional response produced by visible traces of time and decay. Riegl's distinction is important for war-damaged structures, because damage itself becomes a historical layer. A shell-scarred façade documents an event, and it also produces an immediate, felt response in the viewer that a perfectly restored surface cannot.

The Venice Charter (ICOMOS 1964) formalized a cautious approach to intervention. It asked that additions be distinguishable from original fabric and that the contributions of all periods be respected, rather than removed in pursuit of stylistic unity. The Burra Charter (Australia ICOMOS 2013) later broadened the focus from fabric to *cultural significance*, including the associations and meanings a place holds for communities. Taken together, these documents provide a vocabulary for treating damage not simply as a defect to be repaired but as part of what makes a place significant (Jokilehto 1999).

2.2 Adaptive reuse

Adaptive reuse, the process of giving an existing building a new function, has grown from a pragmatic practice into a field of study in its own right. Plevoets and Van Cleempoel (2019) describe it as a discipline located between conservation and design, in which the existing building is treated as a host that conditions, but does not fully determine, the new intervention. Brand (1994) showed that buildings change continuously over their lifetimes, and that the most enduring ones are often those flexible enough to accommodate new uses. For war-damaged structures, adaptive reuse raises a particular question: whether the new use should conceal the damage, accommodate it, or make it central to the building's new identity.

2.3 Resilience

The concept of resilience originated in ecology, where Holling (1973) used it to describe the ability of a system to absorb disturbance and still persist, rather than to return to a single equilibrium state. Applied to cities, resilience is often used in a narrower, engineering sense: the speed of recovery after a shock. This paper follows Holling's broader meaning. In heritage terms, a resilient structure is not necessarily one that is restored to its pre-war condition. It is one that continues to function within the life of the city, physically, socially and symbolically, after its original form or purpose has been lost. The UNESCO Recommendation on the Historic Urban Landscape (2011) points in a similar direction by treating heritage as a dynamic, layered system that must accommodate change.

3. Context: Berlin after 1945

The immediate postwar years in Berlin were dominated by clearance. Enormous volumes of rubble were removed from streets and plots, much of it by hand, and deposited in large mounds at the edge of

the built-up area. Several of these became the city's *Trümmerberge*, or rubble hills, among them the Teufelsberg in the Grunewald forest and the hills of the Volkspark Friedrichshain (Diefendorf 1993).

Reconstruction in the following decades was shaped by the Cold War. In both East and West Berlin, modernist planning ideals encouraged the removal of damaged historic buildings in favour of new development, and many structures that might have been repaired were demolished. At the same time, a small number of ruins were deliberately preserved as reminders of the war (Ladd 1997).

After reunification in 1990, Berlin underwent a renewed and intense debate about its built past. As Huyssen (2003) and Till (2005) have shown, the city became a site where questions of memory, identity and national history were negotiated through architecture. Damaged and fragmentary structures took on new importance in this period, both as evidence of history and as spaces available for new cultural uses. It is within this long and contested history that the transformations discussed below must be understood.

4. Four Modes of Transformation

Examining surviving war-damaged structures in Berlin, four broad modes of transformation can be identified. They are not mutually exclusive, and individual buildings often combine elements of more than one. However, each represents a distinct position on the relationship between damage, use and meaning.

4.1 The conserved fragment: Kaiser Wilhelm Memorial Church

The first mode preserves the damaged structure, or part of it, primarily as a memorial, with new functions accommodated in separate buildings. The Kaiser Wilhelm Memorial Church on the Breitscheidplatz is the best-known example. The church was heavily damaged in an air raid in 1943. Postwar plans to demolish the ruin met strong public resistance, and the eventual solution, designed by Egon Eiermann, retained the damaged west tower and surrounded it with a new church and belfry built between 1959 and 1963.

In this arrangement, the ruined tower no longer performs its original function. It has become a memorial, while the liturgical function moved into the new buildings beside it. The fragment and the new structures work together: the contrast between the broken historic tower and Eiermann's modern, glass-block church gives each its meaning. A smaller example of the same mode is the surviving portico of the Anhalter Bahnhof, which was kept as a free-standing fragment when the rest of the damaged station was demolished. In both cases, resilience lies in the continued symbolic presence of the fragment rather than in any practical use of it.

4.2 Integrated repair: the Neues Museum

The second mode repairs and completes the damaged building while keeping its wounds visible. The Neues Museum on Museum Island, damaged during the war and left as a ruin for several decades, was reopened in 2009 after a reconstruction led by David Chipperfield Architects with the conservation architect Julian Harrap.

The project neither restored the museum to its nineteenth-century appearance nor treated the ruin as untouchable. Surviving historic fabric was conserved in its damaged state, including scarred surfaces

and fragments of decoration. Missing volumes were completed in new materials that follow the original massing but are clearly distinguishable from the old fabric. This approach applies the principles of the Venice Charter at the scale of a whole building. The museum has regained its full function, yet the war remains legible in its walls. Here resilience combines practical recovery with the preservation of memory.

4.3 Programmatic conversion: the Reichsbahnbunker

The third mode gives a war-related structure an entirely new use, with the original structure acting as a host. The Reichsbahnbunker on Reinhardtstraße, a massive air-raid shelter built during the war, illustrates this mode. After 1945 it was used successively as a prison, a storage space for tropical fruit and a club venue. In the 2000s it was acquired by the collector Christian Boros and converted into a private gallery for contemporary art, with a residence added on the roof.

The conversion works with rather than against the bunker's nature. Its thick concrete walls and enclosed rooms suit the controlled environment needed to display art, while openings cut into the structure reveal the depth of the walls. The building's difficult history is not hidden, but it is no longer the primary reason for its existence. This case shows that a structure built for war can be turned to cultural use, though it also raises the question, discussed below, of who gains access to such spaces once they are converted.

4.4 Landscape absorption: the Humboldthain flak tower

The fourth mode absorbs the damaged structure into the landscape. The concrete anti-aircraft towers built in Berlin during the war proved extremely difficult to demolish. At Humboldthain, after partial demolition, the tower was largely covered with rubble and turned into a hill within the park. Part of the structure remains exposed at the top, and its interior can today be visited on guided tours.

Here, the problem of an unwanted and almost indestructible structure was solved by turning it into topography. The same logic applies to the rubble hills more broadly, which turned the debris of destroyed buildings into recreational landscapes. This mode is perhaps the most literal expression of resilience: the material of destruction becomes the ground of new public space. At the same time, it is the mode in which the history of the structure is most easily forgotten, unless it is actively interpreted.

5. Discussion

Read together, the four modes suggest that the adaptive transformation of war-damaged structures in Berlin does not follow a single model. Instead, it forms a spectrum. At one end, the conserved fragment gives priority to memory and gives up use. At the other, landscape absorption gives priority to use and risks losing memory. Integrated repair and programmatic conversion occupy the middle ground, where both are negotiated within a single building.

This spectrum supports the broader understanding of resilience proposed in Section 2. In none of the cases was the original structure returned to its earlier state. What persisted was the structure's capacity to remain part of the city's life and to continue to carry meaning. The Neues Museum is resilient not because it looks as it did in the nineteenth century, but because it combines its restored museum function with a visible record of its history. The Kaiser Wilhelm Memorial Church tower is

resilient because it remains a central landmark and point of reference in the city, even though it no longer serves as a church.

The cases also reveal tensions. The first is between memory and commodification. When a bunker becomes a private gallery, or a ruin becomes a backdrop for tourism, the meaning of the damage can be reduced to atmosphere. The second is between legibility and repair. Each intervention that makes a building usable again risks making its history less visible, which is why the careful distinction between old and new fabric in the Neues Museum has been so widely discussed. The third concerns access and ownership: adaptive reuse can secure the survival of a structure, but it also decides who is able to experience it. Resilience, in other words, is not only a technical or architectural question. It also involves decisions about whose memory is preserved and for whom.

Finally, the Berlin experience has relevance beyond Germany. Cities facing damage from conflict, disaster or climate change, including historic settlements in South Asia, confront similar choices between clearance, restoration and adaptation. The Berlin cases do not provide a model to be copied, as each depends on specific political, economic and cultural conditions. They do, however, show that damage can be treated as a resource for design and memory, rather than only as a loss.

6. Conclusion

This paper has examined how war-damaged structures in Berlin have been transformed rather than erased. It identified four modes of transformation: the conserved fragment, integrated repair, programmatic conversion and landscape absorption. These modes show different ways of balancing use and memory. Across all of them, resilience appears not as a return to an original condition but as the continued ability of a structure to function within the life of the city and to carry meaning.

Further research could extend this typology through a systematic survey of damaged structures across the city, including less prominent examples, and through interviews with users and communities to understand how these transformed places are experienced today. Such work would help to test whether the modes proposed here can inform decision-making for heritage that is damaged or under threat in other urban contexts.

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